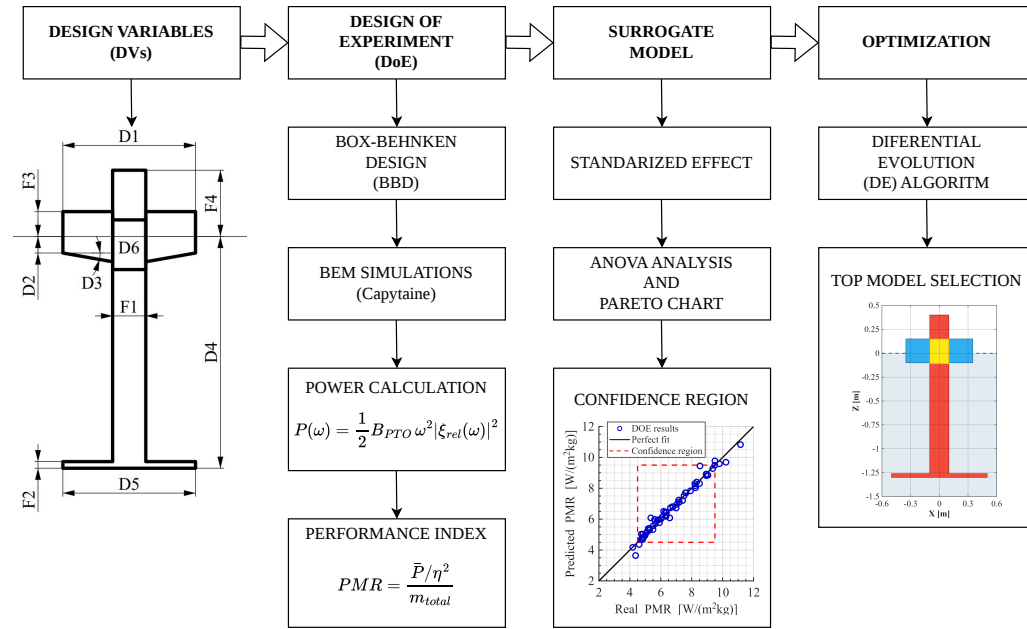


Graphical Abstract

An Integrated Surrogate-Based Optimization Framework for the Preliminary Design of Laboratory-Scale Two-Body Point Absorber Wave Energy Converters

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Highlights

An Integrated Surrogate-Based Optimization Framework for the Preliminary Design of Laboratory-Scale Two-Body Point Absorber Wave Energy Converters

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- Initial development of a tunable Reference Model (RM) for a laboratory-scale Wave Energy Converter (WEC) tailored to the University of New Hampshire (UNH) wave tank constraints. The proposed methodology provides a systematic basis for adapting the design optimization process to the constraints of different laboratory facilities.
- Defined six key Design Variables (DVs) for a laboratory-scale WEC, including five geometric dimensions of the two-body Point Absorber (PA) WEC and the linear damping coefficient of the Power Take-Off (PTO) system.
- Defined the Average Power-to-Mass Ratio (PMR) as the primary performance index, calculated as the ratio of the average absorbed power to the total system mass, enabling fair performance comparisons across laboratory-scale designs.
- Comprehensive Python-based framework for preliminary design optimization of laboratory-scale two-body PA WECs, integrating the open-source Boundary Element Method (BEM) solver CAPYTAIN for hydrodynamic analysis, Box-Behnken (BB) Design of Experiments (DoE) method to efficiently explore the DV space, and quadratic surrogate modeling for high-fidelity response approximation.

An Integrated Surrogate-Based Optimization Framework for the Preliminary Design of Laboratory-Scale Two-Body Point Absorber Wave Energy Converters

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Abstract

This work presents an early-stage parametric design optimization framework for a laboratory-scale two-body Point Absorber (PA) Wave Energy Converter (WEC) tailored to the operational constraints of the University of New Hampshire (UNH) wave tank. Design of Experiments (DoE) approach over six Design Variables (DVs) is employed generating 66 configurations evaluated through Boundary Element Method (BEM) simulations. The Power-to-Mass Ratio (PMR) is selected as the performance metric, normalizing the mean absorbed power by the total structural mass to provide a cost-effective indicator suitable for laboratory-scale prototype comparison. A quadratic surrogate model is fitted to the PMR results of the DoE configurations, which achieves high predictive fidelity. Analysis of Variance (ANOVA) and Pareto analysis identified the floater diameter (D_1) as the dominant individual variable, and the interaction between spar-plate diameter and PTO damping (D_5 – D_6) as the most influential coupled effect. Differential Evolution (DE) algorithm over the surrogate model found 60 candidate configurations that achieve a $\text{PMR} = 9.5$, the upper limit defined for confidence region of the surrogate model, selecting one of them for construction at UNH wave tank,

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expecting a mean absorbed power, normalized by wave amplitude, of 960 [W] with 105 [kg] of overall mass. The proposed framework is open-source and directly transferable to any wave tank facility.

Keywords: Wave Energy Converter (WEC), Point Absorber (PA), Design of Experiments (DoE), Box-Benhken (BB), Hydrodynamic Analysis, Boundary Element Method (BEM), Surrogate Modeling, Laboratory-Scale WEC

1. Introduction

Marine Energy (ME) has emerged as a promising contributor to the decarbonization of the power sector by harnessing the vast energy contained in ocean waves, tides, and currents. In the United States (U.S.), the technical potential of these resources is estimated to satisfy up to 50% of the national electricity demand [1]. Wave Energy Converters (WECs) represent a key class of technologies to extract usable power from ocean wave motion, encompassing diverse operating principles, geometries, and control strategies [2]. Among WECs, Point Absorbers (PAs) have attracted considerable attention due to their compact footprint, omni-directional wave capture, and favorable power-to-size ratio. Single and two-body configurations are actively investigated to exploit relative heave motion, thereby enhancing energy absorption and providing greater flexibility in tuning and control. Recent studies have explored experimental modeling and validation of PA arrays [3], the incorporation of tuned inerters in two-body designs to improve power capture through enhanced inertial effects [4], and the dynamic coupling between floating PAs and catenary mooring systems [5]. However, despite these advances in control strategies, experimental setups and realistic deployment aspects, the systematic influence of the device geometry on hydrodynamic coefficients, system dynamics, resonance behavior, and energy conversion in two-body PAs remains underexplored in a comprehensive, parametric manner—particularly beyond analyses focused on specific control mechanisms or isolated wave conditions.

Although challenges persist in integrating ME systems into the main power grid, in the U.S., WEC development is increasingly aligned with Powering Blue Economy (PBE) applications [6], referred to energy solutions and technological innovations that support ocean-based activities in remote environments often far from the traditional electrical grid (such as aquaculture or

offshore charging stations for autonomous underwater vehicles), which offer a viable near-term pathway for practical deployment of these technologies [6]. Integrating ME technologies, such as WECs, can significantly improve autonomy and sustainability by providing reliable power directly at sea.

A critical step toward improving WECs technologies is the design and development of laboratory-scale prototypes for testing and validation of numerical models. Numerous laboratory-scale WEC prototypes have been developed at academia and national research institutions to enable systematic hydrodynamic characterization, control strategy development, and validation of numerical models. These scaled experiments provide a controlled environment for evaluating device performance, refining design parameters, and reducing technical risk prior to real-scale deployment. The Laboratory Upgrade Point Absorber (LUPA), developed at Oregon State University (OSU), stands as a notable example of an open-source PA device specifically designed for wave tank testing, with its physical design, experimental results, and lessons learned thoroughly documented [7]. The WEC has been experimentally tested under three operating modes, involving floater-only heave motion, coupled floater–spar heave motion, and unrestricted motion [8]. In addition, control co-design strategies have been explored using the open-source WECOpttool to optimize power extraction [9], together with the implementation of both active and passive PTO control strategies [10].

In Europe, several dedicated experimental spaces have been carried out in recent years to study the hydrodynamic behavior of PA arrays. Among these, the WECfarm project has implemented arrays of controllable heaving PAs at Ghent University, providing a valuable framework for the study of WEC farms [11]. Experimental modeling approaches have been developed and validated in this context [12], contributing to a better understanding of the interactions of the array and optimization of performance.

The Reference Model 3 (RM3), developed under Reference Model Project (RMP) of the U.S. Department of Energy (DOE), has emerged as a particularly valuable and widely recognized benchmark for the study of two-body PAs [13]. Originally scaled for grid-connected applications, it has undergone extensive numerical and experimental validation, resulting in a robust, well-documented dataset that is extensively used by the WEC research community [13]. The RM3 device has undergone extensive wave tank testing [14] to validate numerical tools and assess performance under controlled wave conditions, primarily using the open-source software WECSim [15]. However, the RM3 device far exceeds the power requirements of PBE applications.

Therefore, scaling down of it is essential to develop cost-effective and efficient solutions tailored for PBE offshore operations.

In traditional prototyping approaches, laboratory-scale PA WECs are often developed based on a general baseline design, followed by iterative adjustments guided primarily by engineering intuition, parametric studies with limited variables, or trial-and-error testing in wave tanks [16, 17]. Although these methods have provided valuable insights into device hydrodynamics and Power Take-Off (PTO) performance, they typically explore only a narrow subset of the design space and may overlook important interactions among geometric, hydrodynamic, and operational parameters. To overcome these limitations and enable a more systematic and efficient exploration of the design space, the Design of Experiments (DoE) methodology provides a structured framework for identifying key influencing factors, quantifying their main effects and interactions, and optimizing performance with a reduced number of experimental runs compared [18]. Recent applications of DoE in wave energy research have demonstrated its effectiveness in optimizing PA geometries under irregular waves [19], improving system identification for heaving devices [20], and supporting hybrid wind-wave system designs [21].

Despite advances in numerical modeling and full-scale prototyping of PA WECs, persistent gaps still limit efficient preliminary design at laboratory-scale. Comprehensive studies defining suitable ranges for key parameters (such as buoy geometry, mass distribution, and PTO damping) based on international recommendations or aggregated wave tank data remain scarce. Parametric analyses often rely on ad-hoc or device-specific settings, hindering replicability and leading to suboptimal configurations in controlled environments [11, 2]. Moreover, many experimental setups overlook facility-dependent constraints, including water depth, basin geometry, wave generation limits, blockage effects, and evanescent modes, compromising scaling fidelity and performance estimates [8, 11]. In addition, few open-source repositories offer replicable methodologies, raw datasets, or design protocols for laboratory-scale PA WECs, restricting validation, benchmarking, and cumulative progress [8]. Finally, existing guidelines for WEC development lack standardized frameworks specifically adapted to the design of the early-stage laboratory-scale, leaving researchers without clear criteria for geometry selection, PTO configuration, or variable prioritization under constrained experimental conditions [22].

The main novelty of this work is the development of an integrated hy-

hydrodynamic and optimization framework for the preliminary design of a laboratory-scale two-body PA WEC. Hydrodynamic performance is analyzed using a Boundary Element Method (BEM) model, while a Box-Behnken (BB) DoE framework is employed to systematically explore the design space and construct a surrogate model of WEC performance. Six Design Variables (DVs) are defined to study the WEC behavior, where five of them are geometrical dimensions related with the shape of the floating buoy and the spar, and the other one is related with the PTO properties, with bounds selected based on the physical and operational constraints of the UNH wave tank facility, ensuring that all resulting configurations remain experimentally feasible. This preliminary numerical framework is intended to support future research efforts involving scaled experimental testing, enabling validation of the hydrodynamic model and providing further insight into the dynamic response and power capture characteristics of laboratory-scale two-body PA. The primary objective is to establish a tunable WEC RM capable of adapting its dynamic characteristics to enable systematic investigation of hydrodynamic responses and control methodologies under well-defined laboratory conditions. A numerical framework python-based was developed (available on the GitHub ¹ project repository) to efficiently explore the design space of the two-body PA WEC.

The structure of this paper is organized as follows. Section 2 provides the description of a two body PA WEC system. Section 3 focuses on the laboratory-scale PA WEC design, detailing the selection of the DVs and the ranges established for their systematic exploration. Section 4 describes the surrogate-based design and optimization framework, with particular emphasis on the application of DoE methodology. Section 5 presents the simulation and optimization results, selecting one model for construction at UNH wave tank. Finally, Section 6 offers a discussion of the results, followed by conclusions and next steps of the project for future works in Section 7.

2. Two-Body Point Absorber (PA) Wave Energy Converter (WEC) Modeling

This study focuses on a two-body PA WEC design whose geometry is derived from the well-established RM3 device [13] and adapted for labora-

¹<https://github.com/Energy-Control-and-Optimization-Lab/WEC-Tunable-RM/tree/RenewableEnergy2026>

tory scale testing. This section will begin with an overview of the RM3, followed by a description of the numerical modeling framework developed for the proposed two-body PA. The PTO mechanism is modeled as a linear damper characterized by a linear damping coefficient B_{PTO} . This simplification, standard in early-stage WEC design, decouples the hydrodynamic optimization from the complexity of a specific PTO technology, allowing the design space to be explored efficiently while retaining the dominant physics of power extraction through relative heave motion between the two bodies.

2.1. Overview of Reference Model 3 (RM3)

RM3 corresponds to a two-body PA configuration, in which a floating buoy oscillates relative to a submerged spar-plate configuration, in response to ocean wave excitation. The original RM3 system was initially designed at a scale suitable for grid integration. Most of the content in this section is taken from the original report describing the RM3 system [13]. The induced relative motion is harnessed to drive a hydraulic PTO system, which converts the relative motion into hydraulic pressure and drives a fixed-displacement motor and generator to produce electricity [13]. The main dimensions of RM3 are presented in Fig. 1. RM3 was originally designed for the site located off the coast of Eureka, in Humboldt County, California. The sea states at this location exhibit significant wave heights (H_s) ranging from 0.5 [m] to 6 [m], and peak periods (T_p) from 5 [s] to 18 [s], over water depths between 40 [m] and 100 [m]. The rated power output of a single device is 300 [kW], assuming a capacity factor of 30%. Under these conditions, the estimated annual energy production per unit is approximately 700 [MWh].

In the present work, the design focus is on a laboratory-scale two-body PA WEC that preserves the same operating principles as RM3, with a floating buoy oscillating relative to a submerged spar-plate configuration and the relative motion harnessed by a PTO system. The objective is to adapt the RM3 concept to laboratory-scale conditions, maintaining the fundamental hydrodynamic and power take-off behavior while reducing the overall size and power rating for wave tank testing.

2.2. Numerical Modeling of a proposed two-body PA (Hydrodynamics)

For the numerical analysis of our proposed two-body PA, the open-source Python-based BEM solver CAPYTAIN is used, which is formulated within the linear potential wave theory [24, 25]. In this framework, BEM solvers are applied to evaluate the flow around floating bodies in waves by computing

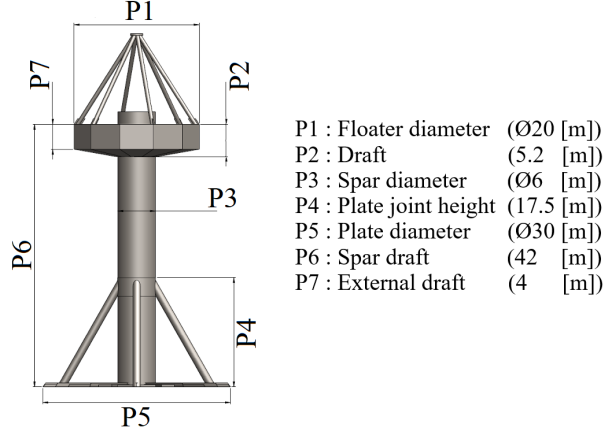


Figure 1: Schematic of Reference Model 3 (RM3) and its main dimensions [23].

the pressure distribution and the associated hydrodynamic loads acting on the structures. Due to the potential flow assumption, the problem reduces to solving the Laplace and Bernoulli equations (Eq. 1 and Eq. 2, respectively) [26]:

$$\nabla^2 \Phi = \frac{\partial^2 \Phi}{\partial x^2} + \frac{\partial^2 \Phi}{\partial y^2} + \frac{\partial^2 \Phi}{\partial z^2} = 0 \quad (1)$$

$$\rho \frac{\partial \Phi}{\partial t} + p + \rho g z = p_0 \quad (2)$$

where Φ is velocity potential, ∇^2 is Laplace operator, ρ is fluid density, p is pressure, p_0 is atmospheric pressure and z is water depth. Eq. 1, and Eq. 2 are solved for the heave Degree of Freedom (DOF) of the floater and the spar (vertical motion). The system of equations is solved by applying the kinematic and dynamic boundary conditions on the free surface and the impermeability boundary condition on the bottom and body surfaces [27]. Additionally, equations of motion written in terms of hydrodynamic coefficients in the frequency domain [26] are presented in Eq. 3 in matrix form.

$$(A(\omega)_{2 \times 2} + M_{2 \times 2})\ddot{\xi}(t)_{2 \times 1} + B(\omega)_{2 \times 2}\dot{\xi}(t)_{2 \times 1} + C_{2 \times 2}\xi(t)_{2 \times 1} = F_e(t)_{2 \times 1} \quad (3)$$

In this equation, A is the added mass matrix, M the body mass matrix, B the damping matrix, C the hydrostatic restoration matrix, F_e the external wave forces vector and t is time. The variable vector ξ represents the 2 DOF of the system, corresponding to the heave motion of each body, while $\dot{\xi}$ and

$\ddot{\xi}$ denote the corresponding velocity and acceleration vectors, respectively. The frequency-domain solution, where ω denotes the wave frequency and i the imaginary unit, is given by Eq. 4. From this solution, the relative heave displacement between the two bodies can be directly derived as $\xi_{rel}(\omega) = \xi_1(\omega) - \xi_2(\omega)$ [26]:

$$\xi(\omega)_{2 \times 1} = [-(M_{2 \times 2} + A(\omega)_{2 \times 2})\omega^2 + B(\omega)_{2 \times 2}\omega i + C_{2 \times 2}]^{-1} F_e(\omega)_{2 \times 1} \quad (4)$$

The total damping acting on the system arises from multiple physical mechanisms. Specifically, B combines three distinct damping components, where $B_h(\omega)_{2 \times 2}$ represents the frequency-dependent hydrodynamic radiation damping, $B_{v2 \times 2}$ corresponds to the linear viscous damping, and $B_{PTO2 \times 2}$ denotes the linear PTO damping assumed as a linear coefficient. The total damping matrix is defined in Eq. 5.

$$B(\omega)_{2 \times 2} = B_h(\omega)_{2 \times 2} + B_{v2 \times 2} + B_{PTO2 \times 2} \quad (5)$$

For the heave-only analysis of the buoy-spar configuration, the total damping matrix $B_{(2 \times 2)}(\omega)$ is presented in Eq. 6.

$$\mathbf{B}(\omega) = \begin{bmatrix} B_{h,b}(\omega) & B_{h,c}(\omega) \\ B_{h,c}(\omega) & B_{h,s}(\omega) \end{bmatrix} + \begin{bmatrix} B_{PTO} & -B_{PTO} \\ -B_{PTO} & B_{PTO} \end{bmatrix} + \begin{bmatrix} B_{v,b} & 0 \\ 0 & B_{v,s} \end{bmatrix} \quad (6)$$

where the hydrodynamic radiation damping component, B_h , is computed using CAPYTAINE, considering the buoy, the spar, and their coupled interaction ($B_{h,b}(\omega)$, $B_{h,s}(\omega)$ and $B_{h,c}(\omega)$ respectively). The PTO damping, B_{PTO} , is assumed constant and represents the linear coupling between the buoy and the spar through the PTO, where the negative off-diagonal terms indicate the opposing forces generated by the relative motion between the two bodies. Finally, the viscous damping matrix, B_v , accounts for energy dissipation due to viscous effects, with $B_{v,b}$ representing the contribution associated with the float, and $B_{v,s}$ corresponding to the spar.

Several assumptions are made about the viscous damping of the two-body system, based on the expected hydrodynamic behavior and the main flow regimes around each component. The buoy is assumed to operate predominantly within a potential-flow regime due to its relatively smooth geometry and moderate heave oscillation amplitudes. Consequently, the viscous damping associated with the buoy, $B_{v,b}$, is considered negligible compared with the corresponding radiation damping term $B_{h,b}$ [28]. In contrast, the spar is subjected to pronounced viscous effects arising from flow separation

and vortex shedding around its submerged plate. Accordingly, the viscous damping contribution of the spar, $B_{v,s}$, is estimated using an empirical relationship [29, 28] given in Eq. 7, which depends on the buoy mass m_b , the natural heave frequency of the buoy $\omega_{0,b}$, and a dimensional empirical viscous damping coefficient associated with the spar C_v , typically ranging between 0.4 and 0.8 [30, 28]. In this work $C_v = 0.6$ is adopted, consistent with the value reported in [28] for a two-body PA WEC of comparable scale.

$$B_{v,s} = 2m_b\omega_{0,b}C_v \quad (7)$$

The time-averaged mechanical power $P(\omega)$ absorbed by the PTO system is calculated from the relative velocity between the buoy and spar, $\dot{\xi}_{rel}(\omega)$, as given in Eq. 8 [31, 32, 33].

$$P(\omega) = \frac{1}{2}B_{PTO}|\dot{\xi}_{rel}(\omega)|^2 \quad (8)$$

For sinusoidal wave excitation, where $\dot{\xi}_{rel}(\omega) = i\omega\xi_{rel}(\omega)$, substituting this expression into Eq. 8 yields $P(\omega)$ as shown in Eq. 9.

$$P(\omega) = \frac{1}{2}B_{PTO}\omega^2|\xi_{rel}(\omega)|^2 \quad (9)$$

This formulation directly relates the power absorption to the relative motion characteristics and PTO damping coefficient, providing the basis for optimization studies and surrogate model development.

2.3. Power Take-Off (PTO) Modeling

The PTO system is modeled as a linear damper, a common simplification in preliminary design stages that retains the essential energy conversion dynamics while remaining compatible with a broad range of PTO technologies. The simplified optimal formulation B_{PTO} is based on the principle of impedance matching for maximum power transfer [34], where the optimal PTO damping per frequency, $B(\omega)_{PTO,opt}$, is related to the potential effective damping ($B(\omega)_{h,eff}$) as presented in Eq. (10), including the impact of viscous damping ($B_{v,total}$) [32]. This formulation matches the PTO load impedance to the resistive components of the system while neglecting the reactive contributions from inertial and stiffness terms, thus providing a computationally efficient and physically intuitive framework for preliminary design optimization. This way, the formulation of $B(\omega)_{PTO,opt}$ is given in Eq. 10.

$$B_{PTO,opt}(\omega) = B_{h,eff}(\omega) + B_{v,total} \quad (10)$$

The effective hydrodynamic damping, $B_{h,eff}(\omega)$, captures the coupled interaction between the bodies, as defined in Eq. (11). The total viscous damping ($B_{v,total}$) accounts for additional energy dissipation mechanisms according to Eq. (12).

$$B_{h,eff}(\omega) = B_{h,b}(\omega) + B_{h,s}(\omega) + 2B_{h,c}(\omega) \quad (11)$$

$$B_{v,total} = B_{v,b} + B_{v,s} \quad (12)$$

Although this represents a simplified approach, it provides sufficient accuracy for exploring the parameter space and offers conservative bounds suitable for preliminary WEC design frameworks.

3. Laboratory-scale Point Absorber (PA) Wave Energy Converter (WEC) Design and Selection of Design Variables (DVs)

This section outlines the overall methodology adopted for the laboratory-scale design of the two-body PA WEC. The section begins with the identification of representative DVs, followed by a detailed assessment of the criteria used to define their feasible ranges. Finally, the appropriate range for each DV is established to ensure consistency with the intended operating and scaling conditions.

3.1. Selection of Design Variables (DVs)

The design is based on the systematic variation of key DVs associated with the buoy and spar configurations. For the buoy, the DVs include the diameter (D_1), the draft (D_2), and the angle of the hull (D_3). The DVs for the spar are the draft (D_4) and the plate diameter (D_5). These geometric DVs define the primary hydrodynamic characteristics of the device and enable the evaluation of different shape-induced performance responses. In parallel, the buoy and spar heights (F_3 and F_4 respectively) remain fixed, as do the spar width (F_1) and the spar plate thickness (F_5). In addition to geometric aspects, B_{PTO} is treated as a DV (D_6), allowing the analysis of how varying damping conditions influence the overall energy absorption efficiency and dynamic behavior of the WEC. An illustration of the PA WEC geometry and its DVs are provided in Figure 2.

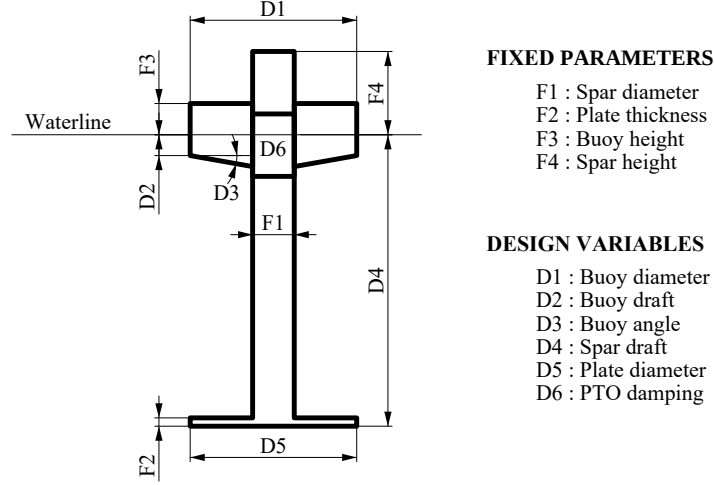


Figure 2: Schematic of laboratory scale two bodies Point Absorber (PA) Wave Energy Converter (WEC) and its Design Variable (DVs).

3.2. Range criterion

The overall dimensions of the two-body PA WEC are constrained by the physical limitations of the UNH wave tank, which measures 3.66 [m] in width, 2.44 [m] in depth, and 36.58 [m] in length. This facility enables controlled experiments involving wave generation at frequencies between 1.6 and 4.0 [rad/s], with wave amplitudes up to 0.25 [m] [35]. These operational boundaries define the feasible scale of the WEC and guide its geometric and dynamic design to ensure consistent and realistic hydrodynamic testing. To determine the DVs range of the proposed WEC, some considerations are carefully taken into account:

1. A key aspect is the ratio between the representative diameters of the model (D_1 and D_5) and the width of the wave tank (W), since an excessively high ratio can lead to boundary effects and unwanted wave reflections with the walls. To ensure the validity of the experimental results, it is recommended that the ratios D_1/W and D_5/W do not exceed 0.3. These guidelines are supported by international organizations such as International Towing Tank Conference (ITTC) [36], Society of Naval Architects and Marine Engineers (SNAME) [37], and Det Norske Veritas (DNV) society of classification [38].
2. The preliminary sizing of D_4 was informed by the dimensional relationships with the depth of the wave tank (h) observed in previous experi-

Table 1: Comparison of dimensions and wave tank facilities from literature, where D_4 is spar draft, and h is tank water depth.

Author	Facility	D_4 [m]	h [m]	h/D_4
Bosma et al. [39]	Oregon State Univ.	2.050	2.74	1.34
Rahimi et al. [40]	Isfahan Univ. of Technology	1.400	2.20	1.57
Beatty et al. [30]	Memorial Univ. of Newfoundland	1.400	2.20	1.57
Huang et al. [41]	Stevens Institute of Technology	1.524	2.40	1.57
Xu et al. [42]	Univ. of California SD	1.200	1.50	1.25
Chen et al. [43]	Shanghai Jiao Tong Univ.	2.044	7.50	3.67

mental studies of similar WEC configurations given in Table 1. Based on the collected database, the maximum recommended ratio between the depth of the wave tank and the spar draft (h/D_4) was established as 1.5, serving as a practical upper bound.

3. The lower limits of the geometric DV ranges were defined on a 1:35 scale for RM3, which is comparable to the 1:33 scale used in the experiments conducted at the University of California, San Diego [42].
4. From preliminary BEM analyses conducted on a reference geometry defined at the center of the geometric DV space, the range of B_{PTO} values was established by evaluating the optimal damping coefficient at each frequency according to Eq. 10, within the operational frequency range of the UNH towing tank, as illustrated in Figure 3.
5. The fixed parameters are defined based on both hydrodynamic and experimental considerations. The spar diameter $F_1 = 0.20$ [m] provides sufficient internal space for the placement and design of the PTO mech-

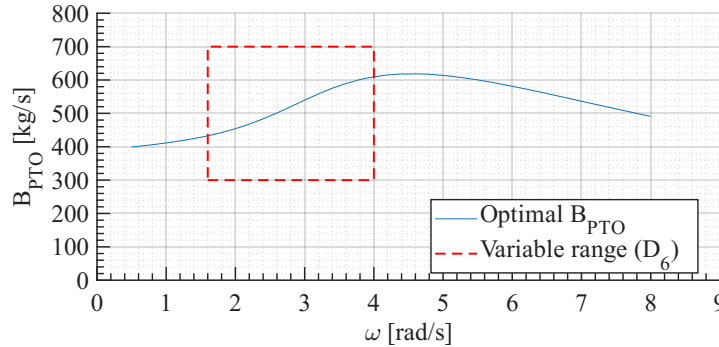


Figure 3: Optimal PTO damping coefficient (B_{PTO}) as a function of frequency (ω) for the mid-range case used in the convergence analysis.

Table 2: Design Variable (DVs) ranges for the laboratory scale two bodies Point Absorber (PA) Wave Energy Converter (WEC).

Variable	Description	Units	Range	
			Min	Max
D_1	Buoy diameter	[m]	0.60	1.00
D_2	Buoy draft	[m]	0.05	0.15
D_3	Buoy angle	[°]	0	20
D_4	Spar draft	[m]	1.20	1.60
D_5	Plate diameter	[m]	0.60	1.00
D_6	PTO damping	[kg/s]	300	700

anism. The spar height $F_2 = 0.40$ [m] and buoy height $F_3 = 0.15$ [m] allow sufficient structural margin for draft adjustments during experimental testing. Finally, the spar-plate thickness $F_4 = 0.04$ [m] was set sufficiently large to ensure adequate panel resolution in the BEM discretization of this region.

3.3. Design Variables (DVs) ranges

Based on the defined constraints and design criteria, the dimensional ranges of the DVs were established to ensure hydrodynamic validity and experimental feasibility within the UNH wave tank facility. The resulting ranges, which serve as the basis for the subsequent numerical simulations, are summarized in Table 2.

4. Surrogate-Based Design and Optimization

The DoE method is constructed over six DVs, based on BB design, generating 66 configurations for which hydrodynamic coefficients are computed via BEM solver CAPYTAINE, and the coupled heave equations of motion are solved in the frequency-domain to obtain spectral power output. A performance index, denoted as the Power-to-Mass Ratio (PMR), is introduced, defined as the wave-amplitude-normalized mean absorbed power divided by the total structural mass, serving as a cost-effectiveness indicator for laboratory-scale prototype comparison. The PMR results obtained for each configuration are used to fit a quadratic surrogate model through Response Surface Methodology (RSM), which is subsequently analyzed via Analysis

of Variance (ANOVA) and Pareto effect decomposition, and then fed into a multi-objective Differential Evolution (DE) optimizer to identify optimal designs.

4.1. Performance index

The performance index PMR is defined as the ratio between the average absorbed power $\bar{P}$ (evaluated within the frequency range of 1.6 to 4.0 rad/s, consistent with the UNH wave tank limits) and the total mass of the system $m_{total} = m_b + m_s$. The hydrodynamic analysis conducted with CAPYTAINE provides dimensionless power values corresponding to a reference regular wave with unit amplitude $\eta = 1[m]$, and this assumption must be explicitly stated when interpreting the numerical results. Under linear potential flow theory, the absorbed power scales quadratically with the wave amplitude, and therefore the numerical results are normalized by the square of the wave amplitude to obtain a power metric expressed in $[W/m^2]$. The final expression of PMR is given in Eq. (13).

$$PMR = \frac{\bar{P}/\eta^2}{m_{total}} \quad [W/(m^2kg)] \quad (13)$$

PMR serves as a normalized performance indicator that enables fair comparison across different geometric configurations while accounting for the trade-off between energy extraction capability and system total mass, making it particularly suitable for laboratory-scale testing where mass constraints and tank dimensions are critical design considerations.

4.2. Design of Experiment (DoE) method

The DoE is a systematic methodology for planning and conducting experiments in order to characterize the relationship between input parameters and system responses with a minimal number of evaluations [18]. By strategically distributing experimental points across the DV space, DoE techniques enable the construction of surrogate models that capture both main effects and coupled interactions, providing a computationally efficient alternative to exhaustive parametric sweeps. This is particularly relevant in WEC design [44] [19], where each evaluation requires the solution of a BEM problem and the subsequent resolution of the frequency-domain equations of motion and power calculation.

Table 3: Design variables and corresponding coded levels used in the Box–Behnken Design.

Variable	Description	Units	Coded level		
			−1	0	+1
D_1	Floater outer diameter	[m]	0.60	0.80	1.00
D_2	Floater draft	[m]	0.05	0.10	0.15
D_3	Hull taper angle	[°]	0	10	20
D_4	Spar column draft	[m]	1.20	1.40	1.60
D_5	Heave-plate diameter	[m]	0.60	0.80	1.00
D_6	PTO damping coefficient	[kg/s]	300	500	700

4.2.1. Box–Behnken (BB) design

The BB design [45], a second-order response surface design belonging to the DoE family, is selected to systematically sample the six DVs. With $k = 6$ factors and $n_c = 6$ center-point replicates, the BB design produced $n = 60 + 6 = 66$ unique experimental configurations, combining factorial points located at the boundaries of the design domain with central points used for curvature estimation. This arrangement enables efficient sampling of the multidimensional variable space while minimizing the number of required BEM computations.

The resulting simulation data are used within a RSM framework [46, 47], widely applied in the optimization of WEC designs [44, 28], to construct a quadratic surrogate model that relate the DVs to the PMR, allowing systematic evaluation and optimization of performance in the design space. All factor levels are linearly coded to the standardized range $[-1, +1]$ for model fitting, allowing direct comparison of DV effects, which is shown in Table 3.

4.2.2. Quadratic surrogate model

A full second-order polynomial is fitted to a response variable y as:

$$\hat{y}(\mathbf{x}) = \beta_0 + \sum_{i=1}^6 \beta_i x_i + \sum_{i=1}^6 \beta_{ii} x_i^2 + \sum_{i < j} \beta_{ij} x_i x_j + \varepsilon, \quad (14)$$

where $\hat{y}$ is the surrogate quadratic model prediction of the performance index PMR; $\mathbf{x} = [x_1, x_2, x_3, x_4, x_5, x_6]^\top \in [-1, +1]^6$ is the coded design vector, with x_i denoting the standardized level of the i -th design variable D_i according to Table 3; β_0 is the overall intercept, representing the predicted response at

the center point of the design space; β_i are the linear coefficients, quantifying the individual main effect of each variable D_i on $\hat{y}$; β_{ii} are the quadratic coefficients, capturing the curvature of the response surface along each design variable axis; β_{ij} are the two-factor interaction coefficients, representing the combined influence of pairs (D_i, D_j) on $\hat{y}$; and ε is the residual error. All coefficients were estimated by ordinary least squares. The model comprises 28 terms in total (1 intercept, 6 linear, 6 quadratic, and 15 interaction terms), which is well-determined given the $n = 66$ experimental runs provided by the BB design.

Surrogate quadratic model adequacy is assessed through four complementary criteria: the coefficient of determination R^2 and its adjusted counterpart R^2_{adj} , which measure the proportion of variance in PMR explained by the model; the mean absolute percentage error (MAPE), which quantifies the average relative deviation between the BEM-computed and surrogate quadratic model predicted values across all 66 experiments; and the ANOVA Lack-of-Fit F -test, which contrasts the model residual variance against the pure error variance independently estimated from the six center-point replicates.

4.2.3. Standardized effects

The relative importance of each term in the surrogate quadratic model (Eq. (14)) is assessed through a Pareto chart of standardized effects. The Pareto chart is a bar diagram in which each bar represents the absolute magnitude of a standardized effect, ordered from largest to smallest. This diagram provides a direct visual ranking of the DVs D_1 – D_6 over the PMR response, their quadratic terms, and their two-factor coupled interactions.

Standardized effects are computed following RSM conventions [47], using the coefficients of the surrogate quadratic model (Eq. (14)) fitted in coded variables (Table 3). The standardized main effect of each DV D_i is defined as $2\beta_i$, representing the change in PMR as x_i varies from -1 to $+1$ while all other variables are held at their center-point values. The standardized interaction effect of a pair (D_i, D_j) is $4\beta_{ij}$, corresponding to the difference in the response change of D_i between the low and high levels of D_j . The standardized quadratic effect of D_i is taken as β_{ii} , capturing the curvature contribution of that variable to PMR.

Statistical significance of each term is assessed by two-sided t -tests on the estimated coefficients at significance level $\alpha = 0.05$, using the pure error variance independently estimated from the six center-point replicates of the BB design as the reference. This approach ensures that the significance of

each effect on PMR is evaluated against a variance estimate that is free from model assumptions, thereby providing an unbiased criterion for identifying the most influential DVs and their interactions.

4.2.4. Optimization

An optimization problem has been solved to maximize the PMR index over the continuous design space $\mathbf{x} \in \mathcal{D}$ using the Differential Evolution (DE) algorithm [48], which is well-suited to non-convex, multi-modal objective functions. The DE approach is selected for solving this optimization with a population size of 500 individuals, 300 generations, crossover probability 0.8, and mutation probability 0.15. To verify robustness of the global optimum and avoid dependence on initial conditions, 60 independent optimization runs were performed using distinct random seeds. The top $N = 60$ candidate designs are extracted from the aggregated run population using a minimum-distance diversity criterion in the design space, ensuring coverage of the Pareto front region rather than convergence to a single point.

5. Simulation Results

This section presents the results of the numerical framework developed for the two-body PA WEC. This section begins by defining a baseline model constructed using the mid-range values of each DV. This model is analyzed to verify the hydrodynamic behavior of the two-body PA system and to establish a reference design for comparison with the optimized models. Then, PMR results for the DoE cases are presented, from which the surrogate quadratic model is constructed. Once validated, a sensitivity analysis based on the Pareto chart of standardized effects quantifies the relative influence of the six DVs and their interactions on PMR. Finally, the surrogate model is used to optimize PMR using a DE algorithm, and the resulting optimal design model is identified and discussed.

5.1. Verification of the Numerical Model of the baseline two-body PA

The baseline two-body PA and its numerical model are evaluated in this section to verify its consistency and ensure it provides a reliable foundation for the subsequent DoE analysis. The baseline model is defined by assigning the midpoint value of each DV range, which represents a geometrically balanced reference case. These discretization parameters were selected based on a mesh convergence study, balancing numerical accuracy with computational

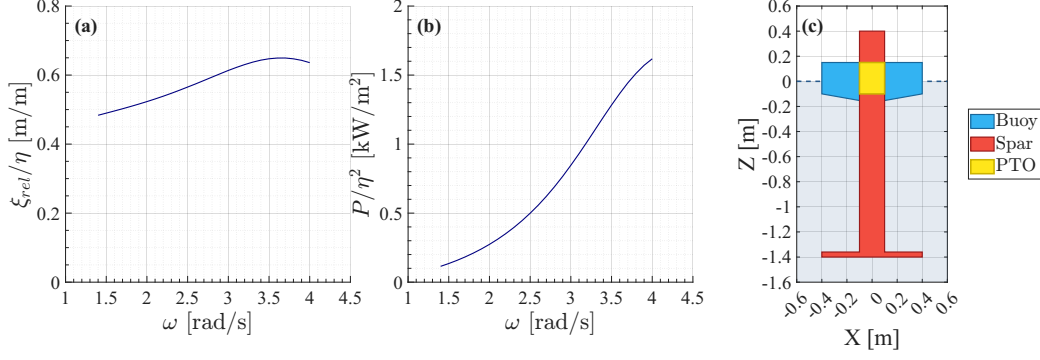


Figure 4: Numerical model results of the two-body PA for the baseline model, where (a) shows the normalized relative motion between the two bodies, (b) the normalized mean absorbed power, and (c) the geometry of the baseline design.

efficiency. The buoy was meshed using 40 circumferential segments and 6 axial subdivisions, while the spar utilized 40 circumferential segments with 50 axial subdivisions and an adaptive radial refinement of 4 intervals across the heave plate thickness ($F2$), to accurately capture the complex flow around this element. The DVs values, total mass and power output information of the baseline model is presented in Table 4.

The numerical model described in Section 2.2 is applied to this baseline model, and the results are presented in Fig. 4. These results include the relative motion between the two bodies (ξ_{rel}), and the mean absorbed power (P), both normalized by the wave amplitude (η). All quantities are evaluated over the UNH wave tank frequency range. The results exhibit smooth and physically consistent trends across the entire frequency band, with no numerical instabilities or anomalous responses, confirming that the model is well-conditioned and suitable to serve as the basis for the parametric study.

Table 4: Design variables and performance metrics for the baseline model. Parameter mapping: D_1 = buoy diameter, D_2 = buoy draft, D_3 = buoy angle, D_4 = spar draft, D_5 = spar plate diameter, D_6 = PTO damping coefficient, m_{total} = Total mass of the system, $\bar{P}/\eta^2$ = Average power in the range of frequencies of UNH wave tank normalized by wave amplitude, PMR = Power-to-Mass Ratio.

D_1 [m]	D_2 [m]	D_3 [°]	D_4 [m]	D_5 [m]	D_6 [kg/s]	m_{total} [kg]	$\bar{P}/\eta^2$ [W/m²]	PMR [W/(m²kg)]
0.800	0.100	10.0	1.400	0.800	500.0	119.43	809.73	6.78

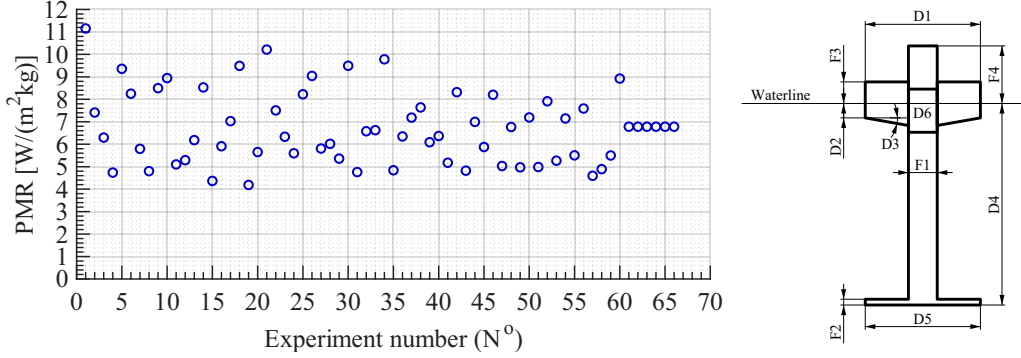


Figure 5: Power-to-Mass Ratio (PMR) results for all 66 Box–Behnken design configurations, showing how the combination of six Design Variables (DVs) affects the primary performance metric of the two-body PA WEC. The complete set of design variable combinations and corresponding results is provided in Appendix A.

5.2. Design of Experiment (DoE) Results

The BB design generated 66 experimental configurations across the six-dimensional design space defined by D_1 – D_6 . For each configuration, a full BEM hydrodynamic analysis was performed, from which the mean absorbed power spectrum were computed, ultimately yielding the PMR value for that design. Figure 5 presents the PMR obtained across all 66 configurations, illustrating the variability of the performance index over the sampled design space and providing the dataset upon which the surrogate quadratic model is constructed.

5.3. Response Surface Model (RSM) Results

The surrogate quadratic model defined in Eq. (14) is evaluated against the 66 BEM simulations for the performance index $\text{PMR} [W/(kg\ m^2)]$. It should be noted that while the model coefficients are estimated in coded variables $x_i \in [-1, +1]$ to enable the standardized effects analysis described in Section 4, the predicted values are back-transformed to physical units for validation purposes, ensuring that the goodness-of-fit metrics R^2 , and MAPE are interpreted in the original scale of the response variable. The complete set of estimated coefficients β_0 , β_i , β_{ii} , and β_{ij} for the surrogate model in physical variables is provided in Appendix B.

$$\begin{aligned}
\text{PMR}(D_1, \dots, D_6) = & \beta_0 \\
& + \beta_1 D_1 + \beta_2 D_2 + \beta_3 D_3 + \beta_4 D_4 + \beta_5 D_5 + \beta_6 D_6 \\
& + \beta_{11} D_1^2 + \beta_{22} D_2^2 + \beta_{33} D_3^2 + \beta_{44} D_4^2 + \beta_{55} D_5^2 + \beta_{66} D_6^2 \\
& + \beta_{12} D_1 D_2 + \beta_{13} D_1 D_3 + \beta_{14} D_1 D_4 + \beta_{15} D_1 D_5 + \beta_{16} D_1 D_6 \\
& + \beta_{23} D_2 D_3 + \beta_{24} D_2 D_4 + \beta_{25} D_2 D_5 + \beta_{26} D_2 D_6 \\
& + \beta_{34} D_3 D_4 + \beta_{35} D_3 D_5 + \beta_{36} D_3 D_6 \\
& + \beta_{45} D_4 D_5 + \beta_{46} D_4 D_6 \\
& + \beta_{56} D_5 D_6 + \varepsilon
\end{aligned} \tag{15}$$

The surrogate model demonstrates excellent predictive accuracy, with $R^2 = 0.978$ and $\text{MAPE} = 2.4\%$. The data points exhibit a strong correlation, yielding the Actual vs. Predicted comparison shown in Figure 6, with the perfect prediction line (black diagonal), indicating robust model performance across the entire range of PMR values. A confidence region is defined between 4.5 and 9.5 $[W/(mkg)]$ (highlighted by the red dashed rectangle), where the surrogate model exhibits optimal linear correlation and minimal prediction scatter. Within this reliable operating range, the model maintains high fidelity with negligible systematic bias, making it particularly suitable for design optimization and parameter sensitivity analyses. Overall, the validated surrogate model provides a computationally efficient surrogate for complex hydrodynamic simulations, enabling rapid exploration of the parameter space for optimal WEC configurations within the established confidence region.

The ANOVA results for the quadratic surrogate model of PMR are reported in Table 5. The model is statistically significant ($F = 63.32$, $p < 0.001$), indicating that the surrogate model in Eq. (14) explains a substantial portion of the observed variability in PMR across the 66 BB design experiments. The model explains 97.83% of the total variance ($R^2 = 0.9783$), with an adjusted coefficient $R_{\text{adj}}^2 = 0.9628$ that accounts for the number of model terms relative to the available degree of freedoms (df), confirming that the high R^2 is not an artifact of overfitting.

Decomposing the model sum of squares by component reveals that both the linear and interaction terms are dominant and statistically significant sources of explained variance ($F = 137.69$, $p < 0.001$ and $F = 57.59$, $p < 0.001$, respectively), with their contributions being comparable in mag-

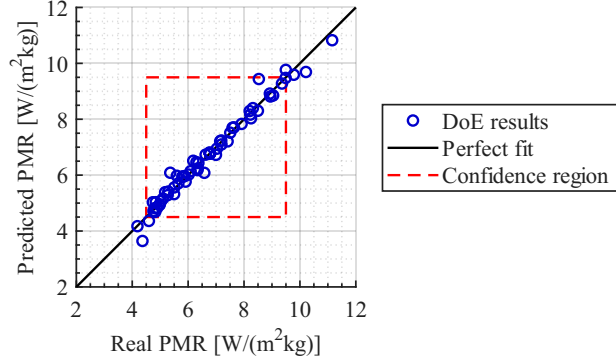


Figure 6: Validation of the quadratic surrogate model for the power-to-mass ratio, showing predicted values from the surrogate model versus results from the original hydrodynamic simulations for all 66 Box–Behnken configurations.

nitude. Unlike the previous analysis, the quadratic terms as a group are also statistically significant ($F = 3.29$, $p = 0.010$), indicating that the PMR response surface exhibits meaningful curvature along at least some of the DV axes. Of the 27 model terms in the Eq. 15, 22 were found individually significant in the overall PMR result, including all 6 linear terms, 5 quadratic terms, and 11 interaction terms. The complete list of significant effects with their standardized values and p -values is provided in Table C.9 in Appendix Appendix C.

5.4. Standardized effects

Figure 7 presents the Pareto chart of standardized effects on PMR, decomposed into three panels: linear, quadratic, and interaction effects. Each

Table 5: Analysis of Variance for the quadratic response surface model of Power-Mass Ratio (PMR). Model statistics: $R^2 = 0.9783$, Adjusted $R^2 = 0.9628$, RMSE = 0.2353 W/(m²kg). SS = Sum of Squares, df = degrees of freedom, MS = Mean Square.

Source	SS	df	MS	F	p -value
Model	94.6683	27	3.506235	63.32	<0.001
Linear	45.7443	6	7.624058	137.69	<0.001
Quadratic	1.0933	6	0.182223	3.29	0.010469
Interaction	47.8307	15	3.188711	57.59	<0.001
Residual	2.1041	38	0.055370		
Total	96.7724	65			

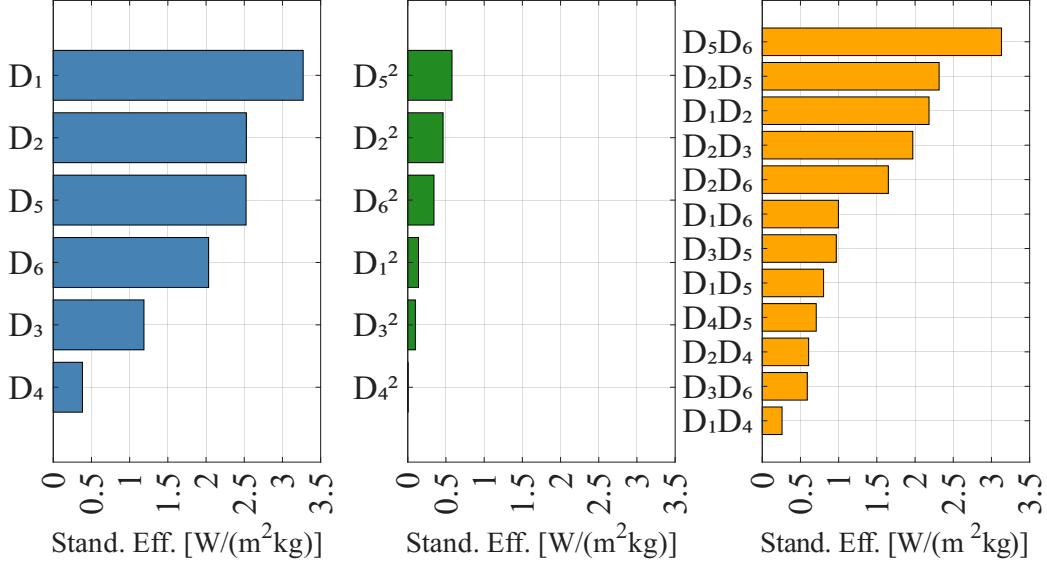


Figure 7: Pareto chart of standardized effects on power-to-mass ratio from response surface methodology analysis. Effects are organized by type: (left) linear effects showing all six Design Variables (DVs), (center) quadratic effects representing curvature, and (right) two-factor interactions showing only statistically significant effects ($p < 0.05$). Parameter mapping: D_1 = buoy diameter, D_2 = buoy draft, D_3 = buoy angle, D_4 = spar draft, D_5 = spar plate diameter, D_6 = PTO damping coefficient.

bar represents how much a given variable or combination of variables influences PMR, with longer bars indicating greater impact.

Among the individual DVs, D_1 has the strongest direct influence on PMR by a notable margin, followed by D_2 and D_5 , which exhibit nearly identical contributions. D_6 ranks fourth, while D_3 and D_4 have comparatively smaller individual effects, with D_4 being the least influential of the six.

The quadratic panel shows that all curvature contributions are small relative to the linear and interaction effects, with D_5^2 being the largest, followed by D_2^2 and D_6^2 . The remaining quadratic terms are negligible in magnitude. This is consistent with the ANOVA result showing that, despite being statistically significant as a group, the quadratic terms contribute substantially less to the total explained variance than the linear and interaction components (Table 5).

The interaction panel includes the largest single effect in the entire chart, with the D_5 – D_6 interaction clearly dominating, confirming that the spar-plate diameter and the PTO damping coefficient must be tuned jointly. The

D_2 – D_5 and D_1 – D_2 interactions are the next most important, followed by D_2 – D_3 and D_2 – D_6 , revealing that D_2 is the variable most involved in coupling effects across the design space. The remaining interactions decrease progressively in magnitude. The full list of significant terms is reported in Table C.9. Overall, the Pareto analysis reveals that PMR is primarily governed by the linear effects of D_1 , D_2 , and D_5 , and by the strong interaction between D_5 and D_6 . While D_1 dominates as an individual variable, the prominence of D_2 across both the linear and interaction panels highlights the floater draft as a critical design parameter whose influence is strongly coupled with the heave-plate diameter and the PTO damping.

5.5. Optimization results

To identify the optimal WEC configuration within the surrogate model, a DE algorithm was employed. This solution corresponds to the global maximum predicted by the surrogate model, with a PMR of 16.69 [$W/(m(^2)kg)$] (geometric parameters, total mass, and power output are summarized in Table 6). BEM simulations, however, yield a lower PMR of 14.11 [$W/(m(^2)kg)$], corresponding to a surrogate prediction error of 18.28%, which is consistent with the uncertainty in PMR for designs located well outside the defined confidence region of the DoE data (Fig. 6). Despite the high predicted performance, the optimal configuration presents practical limitations. The buoy dimensions are at their lower feasible limits, resulting in a particularly compact geometry and lightweight configuration that restricts the integration of the PTO system and other components due to their expected weight and required volume. Moreover, the Pareto chart in Fig. 7 shows that D_4 has a relatively small influence on PMR, suggesting that a reduced spar draft can be adopted with only a minor performance penalty, while increasing flexibility in the design of the mooring system for experimental tests.

This way, according to the confidence region of the DoE results, a target PMR of 9.5 was set as a performance threshold, allowing the extraction of multiple geometrically distinct configurations that satisfy the same performance criterion, preserving flexibility in the final design selection. Using DE optimization algorithm, 60 candidates models were obtained. Figure 8 shows the distribution of the DVs across all 60 candidates considering the coded levels defined in Table 3. Variables D_1 and D_2 tend toward the lower bounds of their design ranges, while D_3 through D_6 shift progressively toward higher values. Additionally, D_4 , D_5 , and D_6 show wider distributions, indicating greater variability in their optimal values.

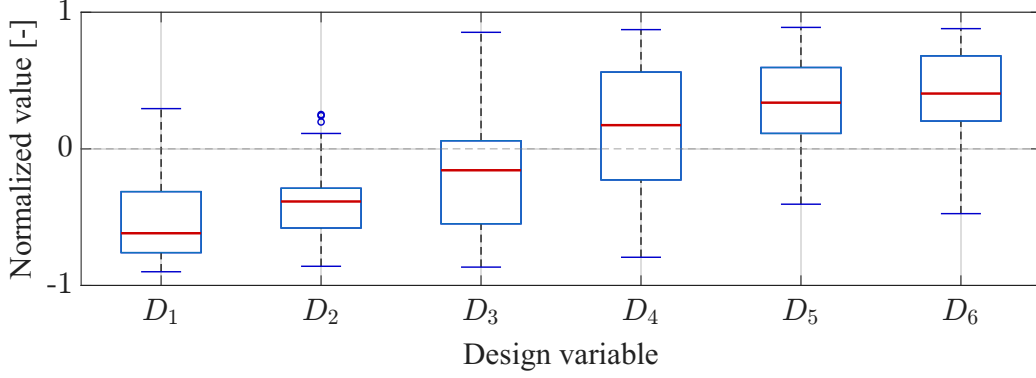


Figure 8: Box plots of the optimal normalized Design Variables (DVs) D_1 through D_6 , obtained from all DE algorithm. The normalized levels are defined according to Table 3. The red line and the blue box indicate the median and the inter-quartile range (IQR), respectively, while the blue lines extend to the most extreme non-outlier values, the outliers are shown as blue circles. Parameter mapping: D_1 = buoy diameter, D_2 = buoy draft, D_3 = buoy angle, D_4 = spar draft, D_5 = spar plate diameter, D_6 = PTO damping coefficient.

The 60 candidate designs are shown in Figure 9, where three were highlighted for further analysis and discussion, being one of them selected for the construction at UNH wave tank. The selection criterion prioritized mass diversity across the design space, ensuring that the chosen configurations represent low, intermediate, and high-mass regions of the candidate set, providing a representative coverage of the explored design space. Table 6 summarizes the corresponding DV values for the models, while Figure 9 presents the normalized average power P/η^2 as a function of the total mass m_{total} for all obtained designs.

The selected model offers a reduced structural mass and a substantial increase in average power production relative to the baseline configuration. These attributes are particularly advantageous for laboratory-scale prototype development, where fabrication costs and material requirements scale directly with device mass. Figure 10 presents a frequency-domain comparison between with the baseline model in terms of relative motion, relative velocity and power absorption. Plot (a) shows the normalized relative displacement ξ_{rel}/η , where the selected model exhibits larger relative motion at almost all frequency range. Plot (b) shows the normalized relative velocity $\dot{\xi}_{rel}/\eta$, which increases monotonically with frequency for both configurations and remains nearly identical across the entire range. Plot (c) shows the normalized absorbed power P/η^2 , where the selected model exhibits a slight increase in power absorption relative to the baseline model across all the

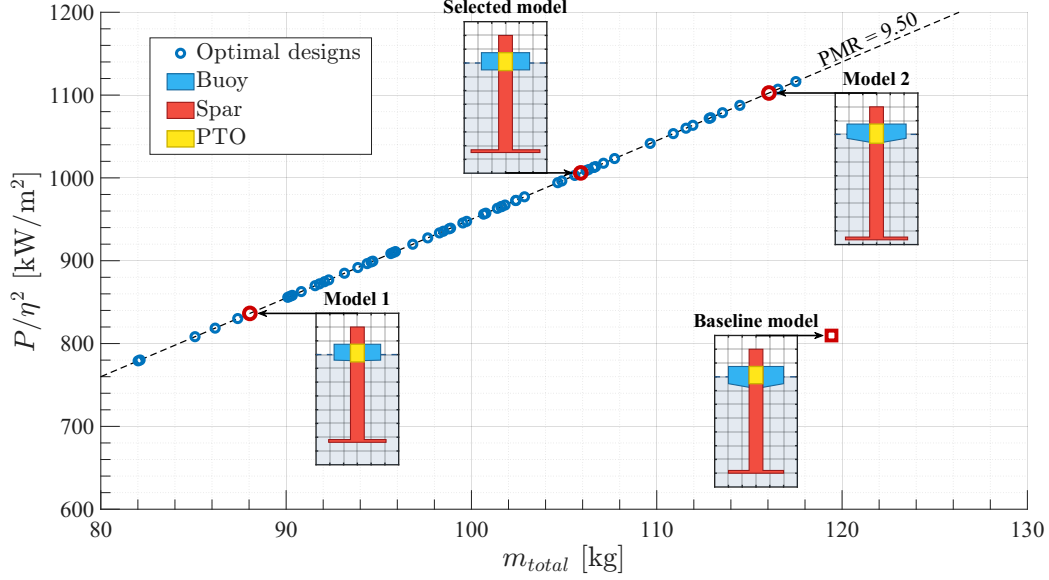


Figure 9: Normalized average power P/η^2 as a function of total mass m_{total} for 60 designs obtained by Differential Evolution (DE) algorithm applied in the surrogate model (blue circles). The dashed line represents the Power-to-Mass Ratio PMR reference of 9.50. Red circles denote the selected top-performing configurations (Model 1, Model 2 and Selected model), while the red square indicates the baseline configuration.

frequency range, despite its lower structural mass.

Table 6: Design variables (DVs) and performance metrics for the baseline model and optimized configurations according the surrogate model of the PMR. The last line shows the global optimal point, which does not follow the confidence region for the DoE. Parameter mapping: D_1 = buoy diameter, D_2 = buoy draft, D_3 = buoy angle, D_4 = spar draft, D_5 = spar plate diameter, D_6 = PTO damping coefficient, m_{total} = Total mass of the system, $\bar{P}/\eta^2$ = Average power in the range of frequencies of UNH wave tank normalized by wave amplitude, $PMR_{predicted}$ = Power-to-Mass Ratio predicted with the surrogate model, PMR_{actual} = Power-to-Mass Ratio obtained by BEM simulations, ϵ = Percentage error of the surrogate model.

	D_1 [m]	D_2 [m]	D_3 [°]	D_4 [m]	D_5 [m]	D_6 [kg/s]	m_{total} [kg]	$\bar{P}/\eta^2$ [W/m²]	$PMR_{predicted}$ [W/(m²kg)]	PMR_{actual} [W/(m²kg)]	ϵ [%]
Baseline model	0.800	0.100	10.0	1.400	0.800	500.0	119.43	809.7	6.78	6.78	0
Model 1	0.676	0.081	1.5	1.274	0.833	524.2	87.6	844.2	9.50	9.64	1.45
Model 2	0.859	0.057	9.8	1.531	0.903	616.6	115.6	1100.9	9.50	9.53	0.31
Selected model	0.700	0.100	0.0	1.300	1.000	540.0	105.9	960.8	9.50	9.07	4.53
Optimal model	0.600	0.050	0.0	1.600	1.000	700.0	92.6	1307.1	16.69	14.11	18.28

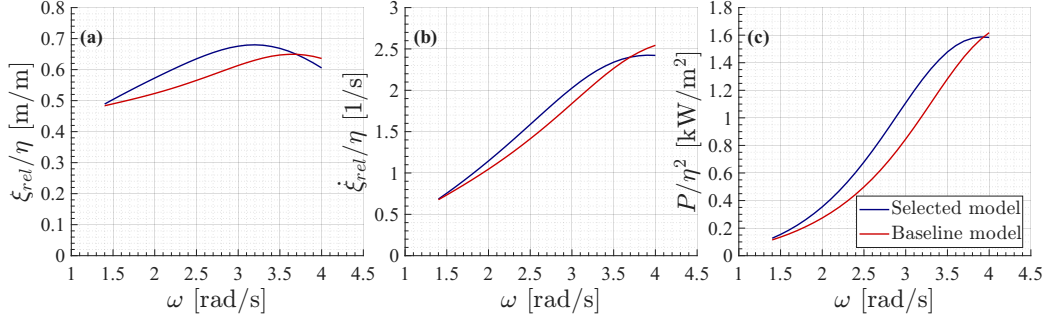


Figure 10: Frequency-domain comparison between the baseline model and selected model. (a) Normalized relative displacement ξ_{rel}/η [m/m], (b) Normalized relative velocity $\dot{\xi}_{rel}/\eta$ [1/s], and (c) Normalized absorbed power P/η^2 [kW/m²], all evaluated over the operational frequency range $\omega \in [1.6, 4.0]$ [rad/s].

6. Discussions

A key contribution of this work is the incorporation of physical constraints into the DV space range from the outset, thereby defining feasible design bounds for the laboratory-scale PA WEC within the early-stage design process. The buoy and spar-plate diameters were constrained by the tank width, the spar draft by the tank depth, and the PTO damping was informed by the preliminary optimal-damping analysis. This ensures that all BEM-evaluated configurations remain physically realizable within the target experimental environment. More importantly, this constraint-embedded strategy provides a replicable framework applicable to any wave tank facility.

Another important insight was the importance of incorporating optimization at the early stages of laboratory device design. By defining PMR as the primary performance metric, the design process enabled the simultaneous consideration of both power output and system mass, rather than focusing exclusively on power production. The analysis of the influence of the DVs on PMR reveals that buoy diameter is the dominant variable, followed by buoy draft and spar-plate diameter, which exhibit nearly identical contributions. At the interaction level, the coupling between spar-plate diameter and PTO damping emerges as the most influential term in the model, reflecting the physical dependence of the optimal PTO damping on the hydrodynamic characteristics of the spar plate. Furthermore, buoy draft appears prominently in multiple significant interaction terms, emerging as the most strongly coupled variable within the design space.

Finally, the optimization results were combined with practical design con-

siderations to select the most suitable configuration for the UNH wave tank. The selected model for construction and laboratory testing exhibits an improved PMR relative to the baseline model, achieving a reduction in structural mass and a slight increase in average power output, while also featuring more manageable physical dimensions that facilitate fabrication and assembly.

7. Conclusions and future work

This study presented a computationally efficient framework for the parametric optimization of a laboratory-scale two-body PA WEC, combining DoE methodology, BEM hydrodynamic analysis, and a surrogate quadratic model to maximize PMR within the operational constraints of the UNH wave tank facility. The six DVs considered in this study were buoy diameter, buoy draft, buoy hull angle, spar draft, spar-plate diameter, and PTO damping. Pareto analysis identified buoy diameter as the dominant single-effect variable, while the interaction between spar-plate diameter and PTO damping emerged as the most influential coupled term in PMR, confirming that the WEC geometry and PTO parameters must be optimized jointly. The results from the confidence region of the DoE, combined with the optimization analysis, led to the selection of an optimal WEC design. The selected configuration achieves a PMR of $9.5 \text{ [W/m}^2\text{kg]}$. The WEC geometry consists of a buoy with a diameter of 0.7 [m] and a draft of 0.1 [m] , coupled with a spar with a draft of 1.3 [m] and a heave plate diameter of 1 [m] . The optimal PTO damping coefficient is 540 [kg/s] . This design is tailored for operation in the UNH wave tank.

The validated surrogate model will serve as the primary tool for defining the final dimensions of the physical WEC prototype to be constructed and tested at the UNH wave tank, enabling rapid performance exploration across candidate designs prior to fabrication. The selected model, characterized by a favorable balance between an increased PMR, reduced structural mass, higher average power output, and dimensions amenable to fabrication, will be constructed following a modular principle. In this approach, the buoy, spar plate, and PTO mechanism will be designed as independently interchangeable components, allowing systematic variation of individual subsystems without requiring reconstruction of the full device.

8. Acknowledgements

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9. Declaration of AI Use in Manuscript Preparation

During the preparation of this work, the author(s) used Claude (Anthropic) and Perplexity AI to assist in improving the clarity, grammar, and overall writing quality of the manuscript. After using these tools, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the accuracy, integrity, and content of the published article.

Appendix A. Design of Experiment (DOE) Results

Table A.7: Complete set of 66 experimental configurations generated by the Box–Behnken design. D_1 : buoy diameter, D_2 : buoy draft, D_3 : buoy angle, D_4 : spar draft, D_5 : spar-plate diameter, D_6 : PTO damping coefficient, m_b : buoy mass, m_s : spar mass, $\bar{P}$: mean absorbed power over $\omega \in [1.6, 4.0]$ rad/s, PMR: power-to-mass ratio.

	D_1 [m]	D_2 [m]	D_3 [°]	D_4 [m]	D_5 [m]	D_6 [kg/s]	m_b [kg]	m_s [kg]	$\bar{P}$ [W]	PMR [W/(m ² kg)]
Exp. 1	0.600	0.050	10.0	1.400	0.800	500	16.19	62.57	878.61	11.15
Exp. 2	0.600	0.150	10.0	1.400	0.800	500	41.22	62.57	769.34	7.41
Exp. 3	1.000	0.050	10.0	1.400	0.800	500	58.14	62.57	759.98	6.30
Exp. 4	1.000	0.150	10.0	1.400	0.800	500	133.23	62.57	927.33	4.74
Exp. 5	0.600	0.100	0.0	1.400	0.800	500	25.03	62.57	819.75	9.36
Exp. 6	0.600	0.100	20.0	1.400	0.800	500	32.62	62.57	785.35	8.25
Exp. 7	1.000	0.100	0.0	1.400	0.800	500	75.09	62.57	797.72	5.79
Exp. 8	1.000	0.100	20.0	1.400	0.800	500	117.60	62.57	865.04	4.80
Exp. 9	0.600	0.100	10.0	1.200	0.800	500	28.71	56.32	722.40	8.50
Exp. 10	0.600	0.100	10.0	1.600	0.800	500	28.71	68.83	872.55	8.95
Exp. 11	1.000	0.100	10.0	1.200	0.800	500	95.68	56.32	775.20	5.10
Exp. 12	1.000	0.100	10.0	1.600	0.800	500	95.68	68.83	870.34	5.29
Exp. 13	0.600	0.100	10.0	1.400	0.600	500	28.71	53.81	510.17	6.18
Exp. 14	0.600	0.100	10.0	1.400	1.000	500	28.71	73.84	874.66	8.53
Exp. 15	1.000	0.100	10.0	1.400	0.600	500	95.68	53.81	652.60	4.37
Exp. 16	1.000	0.100	10.0	1.400	1.000	500	95.68	73.84	1001.87	5.91
Exp. 17	0.600	0.100	10.0	1.400	0.800	300	28.71	62.57	641.45	7.03
Exp. 18	0.600	0.100	10.0	1.400	0.800	700	28.71	62.57	866.15	9.49
Exp. 19	1.000	0.100	10.0	1.400	0.800	300	95.68	62.57	662.50	4.19
Exp. 20	1.000	0.100	10.0	1.400	0.800	700	95.68	62.57	894.35	5.65
Exp. 21	0.800	0.050	0.0	1.400	0.800	500	23.47	62.57	878.88	10.21
Exp. 22	0.800	0.050	20.0	1.400	0.800	500	43.96	62.57	799.75	7.51
Exp. 23	0.800	0.150	0.0	1.400	0.800	500	70.40	62.57	842.10	6.33
Exp. 24	0.800	0.150	20.0	1.400	0.800	500	90.89	62.57	858.75	5.60
Exp. 25	0.800	0.050	10.0	1.200	0.800	500	33.40	56.32	737.77	8.22
Exp. 26	0.800	0.050	10.0	1.600	0.800	500	33.40	68.83	924.27	9.04
Exp. 27	0.800	0.150	10.0	1.200	0.800	500	80.33	56.32	793.58	5.81
Exp. 28	0.800	0.150	10.0	1.600	0.800	500	80.33	68.83	897.64	6.02
Exp. 29	0.800	0.050	10.0	1.400	0.600	500	33.40	53.81	467.43	5.36
Exp. 30	0.800	0.050	10.0	1.400	1.000	500	33.40	73.84	1018.08	9.49
Exp. 31	0.800	0.150	10.0	1.400	0.600	500	80.33	53.81	638.38	4.76
Exp. 32	0.800	0.150	10.0	1.400	1.000	500	80.33	73.84	1014.12	6.58
Exp. 33	0.800	0.050	10.0	1.400	0.800	300	33.40	62.57	635.86	6.63
Exp. 34	0.800	0.050	10.0	1.400	0.800	700	33.40	62.57	938.74	9.78

Continued on next page

Table A.7: Complete set of 66 BBD experimental configurations (continued).

	D_1 [m]	D_2 [m]	D_3 [°]	D_4 [m]	D_5 [m]	D_6 [kg/s]	m_b [kg]	m_s [kg]	$\bar{P}$ [W]	PMR [W/(m ² kg)]
Exp. 35	0.800	0.150	10.0	1.400	0.800	300	80.33	62.57	691.78	4.84
Exp. 36	0.800	0.150	10.0	1.400	0.800	700	80.33	62.57	906.95	6.35
Exp. 37	0.800	0.100	0.0	1.200	0.800	500	46.93	56.32	741.80	7.18
Exp. 38	0.800	0.100	0.0	1.600	0.800	500	46.93	68.83	884.36	7.64
Exp. 39	0.800	0.100	20.0	1.200	0.800	500	67.43	56.32	753.44	6.09
Exp. 40	0.800	0.100	20.0	1.600	0.800	500	67.43	68.83	867.48	6.37
Exp. 41	0.800	0.100	0.0	1.400	0.600	500	46.93	53.81	521.52	5.18
Exp. 42	0.800	0.100	0.0	1.400	1.000	500	46.93	73.84	1004.41	8.32
Exp. 43	0.800	0.100	20.0	1.400	0.600	500	67.43	53.81	584.79	4.82
Exp. 44	0.800	0.100	20.0	1.400	1.000	500	67.43	73.84	988.13	6.99
Exp. 45	0.800	0.100	0.0	1.400	0.800	300	46.93	62.57	643.45	5.88
Exp. 46	0.800	0.100	0.0	1.400	0.800	700	46.93	62.57	898.14	8.20
Exp. 47	0.800	0.100	20.0	1.400	0.800	300	67.43	62.57	653.92	5.03
Exp. 48	0.800	0.100	20.0	1.400	0.800	700	67.43	62.57	879.70	6.77
Exp. 49	0.800	0.100	10.0	1.200	0.600	500	56.86	47.56	519.09	4.97
Exp. 50	0.800	0.100	10.0	1.200	1.000	500	56.86	67.58	895.11	7.19
Exp. 51	0.800	0.100	10.0	1.600	0.600	500	56.86	60.07	582.86	4.98
Exp. 52	0.800	0.100	10.0	1.600	1.000	500	56.86	80.09	1083.73	7.91
Exp. 53	0.800	0.100	10.0	1.200	0.800	300	56.86	56.32	596.02	5.27
Exp. 54	0.800	0.100	10.0	1.200	0.800	700	56.86	56.32	808.81	7.15
Exp. 55	0.800	0.100	10.0	1.600	0.800	300	56.86	68.83	691.91	5.50
Exp. 56	0.800	0.100	10.0	1.600	0.800	700	56.86	68.83	953.92	7.59
Exp. 57	0.800	0.100	10.0	1.400	0.600	300	56.86	53.81	508.59	4.60
Exp. 58	0.800	0.100	10.0	1.400	0.600	700	56.86	53.81	541.23	4.89
Exp. 59	0.800	0.100	10.0	1.400	1.000	300	56.86	73.84	718.52	5.50
Exp. 60	0.800	0.100	10.0	1.400	1.000	700	56.86	73.84	1166.29	8.92
<i>Center-point replicates (Exp. 61–66)</i>										
Exp. 61	0.800	0.100	10.0	1.400	0.800	500	56.86	62.57	809.73	6.78
Exp. 62	0.800	0.100	10.0	1.400	0.800	500	56.86	62.57	809.73	6.78
Exp. 63	0.800	0.100	10.0	1.400	0.800	500	56.86	62.57	809.73	6.78
Exp. 64	0.800	0.100	10.0	1.400	0.800	500	56.86	62.57	809.73	6.78
Exp. 65	0.800	0.100	10.0	1.400	0.800	500	56.86	62.57	809.73	6.78
Exp. 66	0.800	0.100	10.0	1.400	0.800	500	56.86	62.57	809.73	6.78

Appendix B. Surrogate Model Coefficients

Table B.8: Estimated coefficients of the surrogate quadratic model for PMR in physical variables as defined in Eq. (15)

Coefficient	Term	Value
β_0	Intercept	+1.5982
β_1	D_1	-10.0061
β_2	D_2	-27.6011
β_3	D_3	-0.0245
β_4	D_4	-0.3837
β_5	D_5	+24.5213
β_6	D_6	$+6.062 \times 10^{-3}$
β_{11}	D_1^2	+3.5130
β_{22}	D_2^2	+185.0019
β_{33}	D_3^2	$+9.958 \times 10^{-4}$
β_{44}	D_4^2	+0.0711
β_{55}	D_5^2	-14.4795
β_{66}	D_6^2	-8.577×10^{-6}
β_{12}	$D_1 D_2$	+54.5742
β_{13}	$D_1 D_3$	+0.0142
β_{14}	$D_1 D_4$	-1.6181
β_{15}	$D_1 D_5$	-5.0163
β_{16}	$D_1 D_6$	-6.229×10^{-3}
β_{23}	$D_2 D_3$	+0.9853
β_{24}	$D_2 D_4$	-15.1809
β_{25}	$D_2 D_5$	-57.8771
β_{26}	$D_2 D_6$	-0.0413
β_{34}	$D_3 D_4$	-0.0221
β_{35}	$D_3 D_5$	-0.1211
β_{36}	$D_3 D_6$	-7.364×10^{-5}
β_{45}	$D_4 D_5$	+4.4168
β_{46}	$D_4 D_6$	$+1.277 \times 10^{-3}$
β_{56}	$D_5 D_6$	+0.0196

Appendix C. Individual Significant Effects on PMR

Table C.9 lists all statistically significant terms of the surrogate quadratic model ranked by their standardised effect magnitude. The dominant effect is the linear term D_1 , with a standardised effect of 3.2713, followed closely by the D_5 – D_6 interaction (3.1311) and the linear terms D_2 and D_5 (2.5271 and 2.5243, respectively). This result highlights that while the floater outer diameter exerts the greatest individual influence on PMR, the coupling between the heave-plate diameter and the PTO damping coefficient constitutes the single most important interaction in the model, reinforcing the need to tune D_5 and D_6 jointly during the design process.

Table C.9: Significant standardized effects on Power-Mass Ratio (PMR) ($p < 0.05$).

Rank	Effect	Type	Coefficient	Std. Effect	p -value
1	D_1	Linear	−1.6357	3.2713	< 0.001
2	$D_5 D_6$	Interaction	+0.7828	3.1311	< 0.001
3	D_2	Linear	−1.2635	2.5271	< 0.001
4	D_5	Linear	+1.2621	2.5243	< 0.001
5	$D_2 D_5$	Interaction	−0.5788	2.3151	< 0.001
6	$D_1 D_2$	Interaction	+0.5457	2.1830	< 0.001
7	D_6	Linear	+1.0168	2.0337	< 0.001
8	$D_2 D_3$	Interaction	+0.4927	1.9707	< 0.001
9	$D_2 D_6$	Interaction	−0.4126	1.6502	< 0.001
10	D_3	Linear	−0.5936	1.1872	< 0.001
11	$D_1 D_6$	Interaction	−0.2492	0.9966	< 0.001
12	$D_3 D_5$	Interaction	−0.2422	0.9687	< 0.001
13	$D_1 D_5$	Interaction	−0.2007	0.8026	< 0.001
14	$D_4 D_5$	Interaction	+0.1767	0.7067	< 0.001
15	$D_2 D_4$	Interaction	−0.1518	0.6072	< 0.001

Note: Standardised effects computed following RSM conventions. All listed effects are significant at $\alpha = 0.05$. Significance level: $p < 0.001$ for all terms.

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